

Scenario 3: High Illumination Difference

($D > 0.55$)

SIFT/BF

✗ 8% success

5 inliers/tile

LightGlue

△ 38% success

15 inliers/tile

LoFTR

✓ 58% success

Semi-dense

Routing:

LoFTR

Illumination-robust

Scenario 2: Texture-Poor Terrain

($S > 0.65$)

SIFT/BF

✗ 15% success

8 inliers/tile

LightGlue

△ 45% success

18 inliers/tile

LoFTR

✓ 62% success

Semi-dense

Routing:

LoFTR

Detector-free

Scenario 1: Texture-Rich Terrain

($S < 0.35, D < 0.20$)

SIFT/BF

✓ 85% success

45 inliers/tile

LightGlue

✓ 88% success

52 inliers/tile

LoFTR

✓ 82% success

Semi-dense

Routing:

SIFT/BF

Fast classical